

Python Topics Mastered - September 13-27, 2026

Complete Topic List with Difficulty Levels

GITHUB: <https://github.com/MUSTANSIRBM/python-portfolio.git>

BASIC (Beginner Level)

#	Topic	Description
1	Variables & Data Types	Integers, floats, strings, booleans
2	Print & Input Functions	Basic I/O operations
3	String Operations	Concatenation, indexing, slicing
4	Arithmetic Operators	+, -, *, /, %, **, //
5	Comparison Operators	==, !=, <, >, <=, >=
6	Logical Operators	and, or, not
7	If-Else Statements	Conditional logic
8	Elif Statements	Multiple conditions
9	For Loops	Iteration over sequences
10	While Loops	Conditional iteration
11	Break & Continue	Loop control statements
12	Range() Function	Number sequence generation

INTERMEDIATE

#	Topic	Description
13	Lists	Creation, indexing, slicing, methods
14	List Comprehension	Concise list creation
15	Tuples	Immutable sequences
16	Dictionaries	Key-value pairs, methods
17	Dictionary Comprehension	Concise dict creation
18	Nested Data Structures	Lists inside dicts, nested dicts
19	Functions	Definition, parameters, return
20	Default Arguments	Functions with default values
21	*args & **kwargs	Variable-length arguments
22	Lambda Functions	Anonymous functions
23	Map() Function	Applying functions to iterables
24	Filter() Function	Filtering iterables
25	Zip() Function	Combining iterables
26	Enumerate() Function	Index-value pairs
27	Sorting with Lambda	Custom sorting logic
28	Classes & Objects	OOP fundamentals
29	init Method	Constructor
30	str Method	String representation
31	Instance Methods	Object behavior

32	Instance Attributes	Object properties
33	File I/O - Read	open(), read(), readlines()
34	File I/O - Write	open(), write()
35	File I/O - Append	open() with 'a' mode
36	File I/O - Read & Write	open() with 'r+' mode
37	strip(), split()	String processing
38	Exception Handling	try, except, finally
39	Importing Modules	import, from...import
40	datetime Module	Timestamps, formatting
41	JSON Module	json.dump(), json.load()

ADVANCED

#	Topic	Description
42	Inheritance	Parent-child classes
43	super() Function	Calling parent methods
44	Method Overriding	Redefining parent methods
45	Encapsulation	Private attributes
46	Object Composition	Objects as attributes
47	Decorators	@ syntax, function wrapping
48	Custom Decorators	Creating custom decorators
49	Decorator with Parameters	Advanced decorators

50	os Module	File system operations
51	shutil Module	File copy/move operations
52	json.dumps() / json.loads()	JSON string conversion
53	Complex Nested Loops	Pattern printing, matrix operations
54	ETL Implementation	Extract-Transform-Load patterns
55	Data Persistence	Saving/Loading objects to files
56	Business Logic	Real-world application logic



DIFFICULTY SUMMARY

Level	Topics Count
● Basic	12
● Intermediate	29
● Advanced	15
Total	56 Topics



MAJOR PROJECTS THAT DEMONSTRATE THESE TOPICS

Project	Topics Used
ATM System	Classes, Methods, File I/O, Validation
Task Manager	Classes, Lists, File I/O, Status Management
Student Management	Classes, Aggregation, Data Analysis

Hotel Booking	Classes, State Management, Search Logic
Train Booking	Nested Dicts, Business Logic, Pricing
Project Management	Inheritance, Decorators, JSON
University Management	Inheritance, Decorators, JSON, Analytics
My_Sorter	os, shutil, File Operations, Automation

KEY SKILLS HIGHLIGHT FOR SENIOR

-  **Object-Oriented Programming** (56% of topics)
-  **Data Persistence** (ETL - All projects)
-  **File System Operations** (os, shutil)
-  **Custom Decorators** (2 projects)
-  **JSON Processing** (2 projects)
-  **Complex Data Structures** (Nested dicts/lists)
-  **Business Logic Implementation** (14 projects)

Total Topics Mastered: 56

Projects Built: 16

Python Libraries Used: 7

"From basic loops to custom decorators - complete Python proficiency achieved in 14 days!" 🏆